

Monte Carlo Options Pricing System

M6 Checkpoint: FastAPI Backend

1 Architecture

Atomic endpoints—each maps to exactly one module (M1–M5). The client orchestrates the workflow:

`/calibrate → /simulate → /price-option → /risk/*`

No business logic in the API layer; pure delegation plus response shaping. `POST /data/fetch` omitted—ingestion is always offline.

2 Endpoints

Route	Method	Module	Notes
<code>/health</code>	GET	—	Health check
<code>/data/tickers</code>	GET	M1 storage	List available tickers
<code>/data/{ticker}</code>	GET	M1 storage	OHLCV with optional date range
<code>/calibrate</code>	POST	M2 calibration	Logs to MLflow
<code>/simulate</code>	POST	M3 simulation	Returns bands + sample paths
<code>/price-option</code>	POST	M4 pricing	Price + Greeks + BS benchmark; logs to MLflow
<code>/risk/metrics</code>	POST	M5 risk	Single-asset risk metrics
<code>/risk/portfolio</code>	POST	M5 risk	Multi-asset correlated simulation
<code>/risk/scenarios</code>	GET	M5 scenarios	List preset scenario names
<code>/mlflow/experiments</code>	GET	MLflow	Lightweight experiment listing

3 Discriminated Union for Model Params

Added `model_type: Literal[...]` with defaults to each calibration params model:

`GBMParams("gbm"), JumpDiffusionParams("jump_diffusion"), HestonParams("heston")`

`ModelParams = Annotated[..., Discriminator("model_type")]`

Defaults ensure all existing constructors remain unbroken.

Related fix. `CalibratorBase.goodness_of_fit` now excludes `model_type` from k in the AIC/BIC penalty:

$$\text{AIC} = -2\ell + 2k, \quad \text{BIC} = -2\ell + k \ln n$$

Without the fix, every model’s information criteria would have been silently inflated by a non-parameter tag field.

4 Async Strategy

`async def` with explicit `asyncio.to_thread()` for compute-heavy endpoints (calibrate, simulate, price, risk). This keeps the event loop unblocked and makes the async boundary visible rather than relying on FastAPI's implicit threadpool for plain `def` endpoints.

5 Response Shaping

5.1 Simulation Response

Percentile bands (p_5 , p_{25} , p_{50} , p_{75} , p_{95}) computed at each time step, then downsampled to ~ 50 points via simple stride (every N -th element, always including first and last). Five random sample paths included, also downsampled.

5.2 Pricing Response

Payoff histogram via `np.histogram(payoffs, bins=50)`, returned as `{bin_edges, counts}`. Requires a separate simulation run because `MCPricer.price()` returns the scalar discounted mean, not the payoff sample. With a seed, this reproduces the priced paths exactly. Overhead: $\sim 10\%$ on top of the ~ 10 simulations Greeks already cost.

5.3 Risk Response

VaR and CVaR keyed by confidence level strings ("95", "99"). Max drawdown reports both mean and 95th percentile. Raw P&L sign convention (negative = loss) from M5 preserved.

6 Physical vs Risk-Neutral Measure

The API layer preserves the measure distinction established in M4/M5:

Endpoint	Measure
<code>/simulate</code> , <code>/risk/metrics</code> , <code>/risk/portfolio</code>	Physical: $\mu = \mu_{\text{calibrated}}$
<code>/price-option</code>	Risk-neutral: $\mu = r - q$

7 MLflow Logging

- `/calibrate`: experiment `calibration/{ticker}`, logs params (ticker, model_type, look-back_days) and fit metrics.
- `/price-option`: experiment `pricing/{option_type}`, logs params (option_type, S_0 , K , T , r , model_type) and metrics (price, std_error, Greeks).
- Graceful degradation: if MLflow is unavailable, proceed with `mlflow_run_id = None`.
- `/simulate` and `/risk/*` do not log to avoid noise.

8 Error Handling

Uniform `ErrorResponse(error, detail, context)` shape for all errors:

Exception	HTTP Status
ValueError	422
KeyError (unknown ticker/model)	404
SimulationTooLargeError ($n > 100,000$)	413
RequestValidationError (Pydantic)	422

The 413 is enforced in routers, not Pydantic validators, because Pydantic's `le` constraint can only produce 422.

9 Implementation Judgment Calls

- (a) **api/utils.py added.** Houses `downsample`, `compute_histogram`, `load_recent_closes`, and the MLflow helper. Not in the spec's file list, but these functions needed a home that wasn't `schemas.py` or `dependencies.py`. `load_recent_closes` is shared by `/calibrate` and `/risk/portfolio`.
- (b) **r added to risk requests.** `r: float = 0.0` on `RiskMetricsRequest` and `PortfolioRiskRequest`. The Sharpe ratio formula $\frac{E[r]-r_f}{\sigma[r]}$ requires a benchmark rate. It defaults to zero, never reaches a simulator, and the endpoints remain on the physical measure.
- (c) **K made optional.** `K: float | None` because floating-strike lookbacks have no strike. A model validator requires K for all other payoffs and `barrier` for exactly the knock-out types.
- (d) **Alphabetical sort before `simulate_portfolio`.** `estimate_correlation_matrix` returns its matrix in sorted-ticker order. Without re-sorting the asset list to match, the correlation assignments would be silently wrong.
- (e) **goodness_of_fit parameter count fix.** The new `model_type` field would have inflated k in AIC/BIC. Now excluded from the count.

10 Test Summary

Scope	Tests
M1–M5 (pre-existing, non-DB)	345
M1 DB tests	28
M6 API tests	109
Total passing	482

4 live-yfinance tests deselected. All prior tests unchanged.

11 Files

```
src/api/
  __init__.py
  main.py
  schemas.py
  dependencies.py
  utils.py
routers/
  __init__.py
  data.py
```

```
calibration.py
simulation.py
pricing.py
risk.py
mlflow_runs.py
```

Dependencies added: `fastapi`, `uvicorn[standard]`, `httpx (TestClient)`, `mlflow`.

12 Cumulative Milestone Status

Milestone	Status	Cumulative Tests
M1 — Data Ingestion	✓	—
M2 — Calibration Engine	✓	24
M3 — Simulation Engine	✓	96
M4 — Options Pricing	✓	183
M5 — Risk Analytics	✓	345
M6 — FastAPI Backend	✓	482

Next: M7 — React Frontend Dashboard